Programming Languages and Computation

Week 8: Partial Correctness & Loop Invariants

Hoare Logic

* 1. Consider the following *invalid* Hoare triple: $\{x \leq y\}\ y \leftarrow y * 2\ \{x \leq y\}$. Find an initial state $\sigma \in \text{State}$ and a trace from the configuration $\langle y \leftarrow y * 2, \sigma \rangle$ that contradicts this triple.

* 2. Find a statement S such that $\{\text{true}\}\ S\ \{z \leq x\ \&\&\ z \leq y\}$.

** 3. We write $d \mid n$ as shorthand for the assertion $\exists k. n = k * d$; that is, n is a multiple of d, or equivalently *d divides n*. For example, $y \mid x$ is satisfied by those states in which the value of x is a multiple of the value of y.

For each of the following statements, compute the weakest pre-condition required to ensure the post-condition $(y \mid x)\ \&\&\ y \geq 0$ holds after execution:

(a) $x \leftarrow x * x$

(b) $\text{if } y \leq x\ \{x \leftarrow x - y\}\ \text{else}\ \{y \leftarrow y - x\}$

** 4. Using your answers to the previous question, for each of the following Hoare triples determine whether they are valid or not. You should justify your answer.

(a) $\{(y \mid x)\ \&\&\ y \geq 0\}\ x \leftarrow x * x\ \{(y \mid x)\ \&\&\ y \geq 0\}$

(b) $\{(y \mid x)\ \&\&\ y \geq 0\}\ x \leftarrow x * x\ \{x \geq y\}$

(c) $\{(y \mid x)\ \&\&\ y \geq 0\}\ \text{if } y \leq x\ \{x \leftarrow x - y\}\ \text{else}\ \{y \leftarrow y - x\}\ \{(y \mid x)\ \&\&\ y \geq 0\}$

** 5. Suppose $S_1, S_2, S_3 \in \mathcal{S}$ are statements satisfying the following Hoare triples:

- $\{x \leq y\}\ S_1\ \{y \leq x\ \&\&\ x \leq 0\}$
- $\{\forall z. x * z = 0\}\ S_2\ \{y = z\}$
- $\{x > y\ \&\&\ z = x\}\ S_3\ \{y \leq x\}$

Then which of the following triples can be concluded from these? You should briefly justify your answer, e.g. by strengthening pre-conditions, weakening post-conditions, and combining triples.

(a) $\{x \leq y \;\&\&\; x = 0\}\ S_1;\ S_2\ \{y = z\}$

(b) $\{z = x\}$ if $x \leq y$ $\{S_1\}$ else $\{S_3\}$ $\{y \leq x\}$

(c) $\{x \leq y\}\ S_1;$ if $y = 0$ $\{S_2\}$ else $\{y \leftarrow z\}$ $\{y = z\}$

** 6. Let $\mathsf{cmp}_{i,t,f}$ be some statement whose behaviour is determined as follows:

- CMP1. If $\sigma(i) = \sigma(t)$, then $\langle \mathsf{cmp}_{i,t,f}, \sigma \rangle \to^* \sigma[f \mapsto 1]$;
- CMP2. Otherwise, if $\sigma(i) \neq \sigma(t)$, then $\langle \mathsf{cmp}_{i,t,f}, \sigma \rangle \to^* \sigma[f \mapsto 0]$.

Let T be the statement $\mathsf{cmp}_{i,t,f};\ i \leftarrow i + 1$.

(a) Determine the state σ such that $\langle T, [i \mapsto 2, t \mapsto 2, f \mapsto 0] \rangle \to^* \sigma$, justifying your answer in relation to CMP1. and/or CMP2.

(b) Calculate $\mathsf{WP}(T, f = 1 \parallel i \leq t)$ and simplify it as much as you can.

(c) Using your answer to the previous question, determine whether each of the following triples is valid.

$$\{i \leq t\}\ T\ \{f = 1 \parallel i \leq t\} \qquad\qquad \{\mathsf{true}\}\ T\ \{f = 1 \parallel i \leq t\}$$

*** 7. Strengthening pre-conditions and weakening post-conditions allows us to adapt a Hoare triple to a more general context. However, it is also useful to be able to adapt Hoare triples to contexts with other unrelated variables. This can be done through the *constancy rule*:

$$\frac{\{P\}\ S\ \{Q\}}{\{P \;\&\&\; R\}\ S\ \{Q \;\&\&\; R\}}\ \mathsf{FV}(R) \cap \mathsf{mod}(S) = \emptyset$$

Where $\mathsf{FV}(P)$ for a predicate $P \subseteq \mathsf{State}$ is defined as the set of variables $x \in \mathsf{Var}$ such that $\sigma \in P$ but $\sigma[x \mapsto n] \notin P$ for some state $\sigma \in \mathsf{State}$ and $n \in \mathbb{Z}$; intuitively, those variables whose value effect whether a state is in the set of not and $\mathsf{mod}(S)$ is the set of variables appearing anywhere in the program.

(a) Suppose that S_1 and S_2 are two statement such that $\{P_1\}\ S_1\ \{Q_1\}$ and $\{P_2\}\ S_2\ \{Q_2\}$ where:
- $\mathsf{FV}(P_1), \mathsf{FV}(Q_1) \subseteq \mathsf{mod}(S_1)$;
- $\mathsf{FV}(P_2), \mathsf{FV}(Q_2) \subseteq \mathsf{mod}(S_2)$;
- And $\mathsf{mod}(S_1) \cap \mathsf{mod}(S_2) = \emptyset$.

Justify how $\{P_1 \;\&\&\; P_2\}\ S_1;\ S_2\ \{Q_1 \;\&\&\; Q_2\}$ can be derived from the constancy rule.

(b) Suppose that $P \subseteq \mathsf{State}$ is some set of states. Prove that, if $\sigma \in P$ and $\sigma(x) = \sigma'(x)$ for all $x \in \mathsf{FV}(P)$ and $\#\{x \in \mathsf{Var} \mid \sigma(x) \neq \sigma'(x)\}$ is finite, then $\sigma' \in P$. You may wish to prove it by induction on $\#\{x \in \mathsf{Var} \mid \sigma(x) \neq \sigma'(x)\}$.

(c) Prove that the rule of constancy holds that is:

$$\{P\}\ S\ \{Q\} \Rightarrow \{P\ \&\&\ R\}\ S\ \{Q\ \&\&\ R\}$$

for any statement S, and predicates $P, Q, R \subseteq \mathsf{State}$ such that $\mathsf{FV}(R) \cap \mathsf{mod}(S) = \emptyset$. You may wish to use a result from the previous worksheet.

Loop Invariants

Recall that an assertion I is a *loop invariant* for while e $\{S\}$ if $\{I\ \&\&\ e\}\ S\ \{I\}$, i.e. $I\ \&\&\ e \subseteq \mathsf{WP}(S, I)$. To show that $\{P\}$ while e $\{S\}$ $\{Q\}$ holds using an invariant I, we additionally need $P \subseteq I$, i.e. the invariant implies the pre-condition, and $I\ \&\&\ !\,e \subseteq Q$, i.e. the invariant implies the post-condition on exit.

** 8. For each of the following statements, determine whether $q \leq x$ is a loop invariant. If so, justify why. Otherwise, provide a counter-example.

(a) while $y * q \leq x$ $\{q \leftarrow q + 1\}$

(b) while $!(q \leq x)$ $\{x \leftarrow x - 1\}$

(c) while $!(x \leq q * q)$ $\{q \leftarrow q + 1\}$

** 9. Find a loop invariant that can be used to demonstrate the following Hoare triple. You should justify your solution by checking that the invariant is implied by the pre-condition (after the initialisation), preserved by the loop body, and implies the post-condition, calculating weakest pre-conditions as necessary.

$$\begin{array}{l}\{0 \leq x\ \&\&\ 0 < y\}\\ q \leftarrow 0;\\ r \leftarrow x;\\ \text{while } y \leq r\ \{\\ \quad r \leftarrow r - y;\\ \quad q \leftarrow q + 1\\ \}\\ \{x = q * y + r\ \&\&\ r < y\}\end{array}$$

*** 10. Find a loop invariant that can be used to demonstrate the following Hoare triple. You should justify your solution by checking that the invariant is implied by the pre-condition (after the initialisation), preserved by the loop body, and implies the post-condition, calculating weakest pre-conditions as necessary.

$$
\begin{array}{l}
\{n \geq 0\} \\
i \leftarrow 0; \\
s \leftarrow 0; \\
\text{while } !(i = n)\ \{ \\
\quad s \leftarrow s + 2 * i + 1; \\
\quad i \leftarrow i + 1 \\
\} \\
\{s = n * n\}
\end{array}
$$

*** 11. Let $\mathsf{cmp}_{i,t,f}$ be the statement from the earlier exercise, satisfying CMP1. and CMP2. Consider the following program, which we shall refer to as S, that searches for the value of t amongst the numbers from 1 up to n.

$$
\begin{array}{l}
i \leftarrow 1; \\
f \leftarrow 0; \\
\text{while } i \leq n \;\&\&\; f = 0\ \{ \\
\quad \mathsf{cmp}_{i,t,f}; \\
\quad i \leftarrow i + 1 \\
\}
\end{array}
$$

(a) Determine the final state σ such that $\langle S, [n \mapsto 5, t \mapsto 3] \rangle \rightarrow^* \sigma$ and justify your answer in relation to CMP1. and/or CMP2.

(b) Find a loop invariant that justifies the following partial correctness property and prove that it is valid by showing that it is implied by the pre-condition (after the initialisation), preserved by the loop body, and implies the post-condition.

$$\{1 \leq t \;\&\&\; t \leq n\}\ S\ \{f = 1\}$$

You should make reference to CMP1. and CMP2. in your proof.

*** 12. Find a pair of loop invariants (one for each loop) that together can be used to demonstrate the following Hoare triple. You should justify your solution by checking, for each loop, the three constraints on a loop invariant, calculating weakest pre-conditions as necessary.

$$
\begin{array}{l}
\{0 \leq n\} \\
i \leftarrow 0; \\
j \leftarrow 0; \\
\text{while } i \leq n - 1\ \{ \\
\quad j \leftarrow 0; \\
\quad \text{while } j \leq i\ \{ \\
\quad\quad j \leftarrow j + 1 \\
\quad \}; \\
\quad i \leftarrow i + 1 \\
\} \\
\{j = n\}
\end{array}
$$

** 13. Find a loop invariant for the following program that demonstrates that this program has at least one non-terminating trace. You should justify your answer in relation to the definition of a Hoare triple.

$$\text{while } x > 0 \ \{x \leftarrow x + y\}$$